AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 20 · STUDENT LESSON

Mean, Median, and Baselines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Fit from training. Freeze the constant. Score new targets.

Compare mean and median numeric baselines without leaking evaluation or deleting unexplained extremes.

Keep one habit: Complete fitting example → pictured held-out tie → fixed-target tournament → missing/extreme probes → independent new constants.

MAIN LESSON · SECTION 1

What does a simple constant predict for every new case?

> Fit the constant on training targets and compare it on separate targets.

Learning goals

I can fit **mean** and **median constant baseline**s on **training** targets.

I can compare constant **prediction**s on separate records.

I can keep fitted constants independent of evaluation answers.

Recall signed-error cancellation

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.
- Why can signed residuals hide error size?

End goal

By the end, I can explain **mean**, **median**, and baselines using a traced example and a stated limitation.

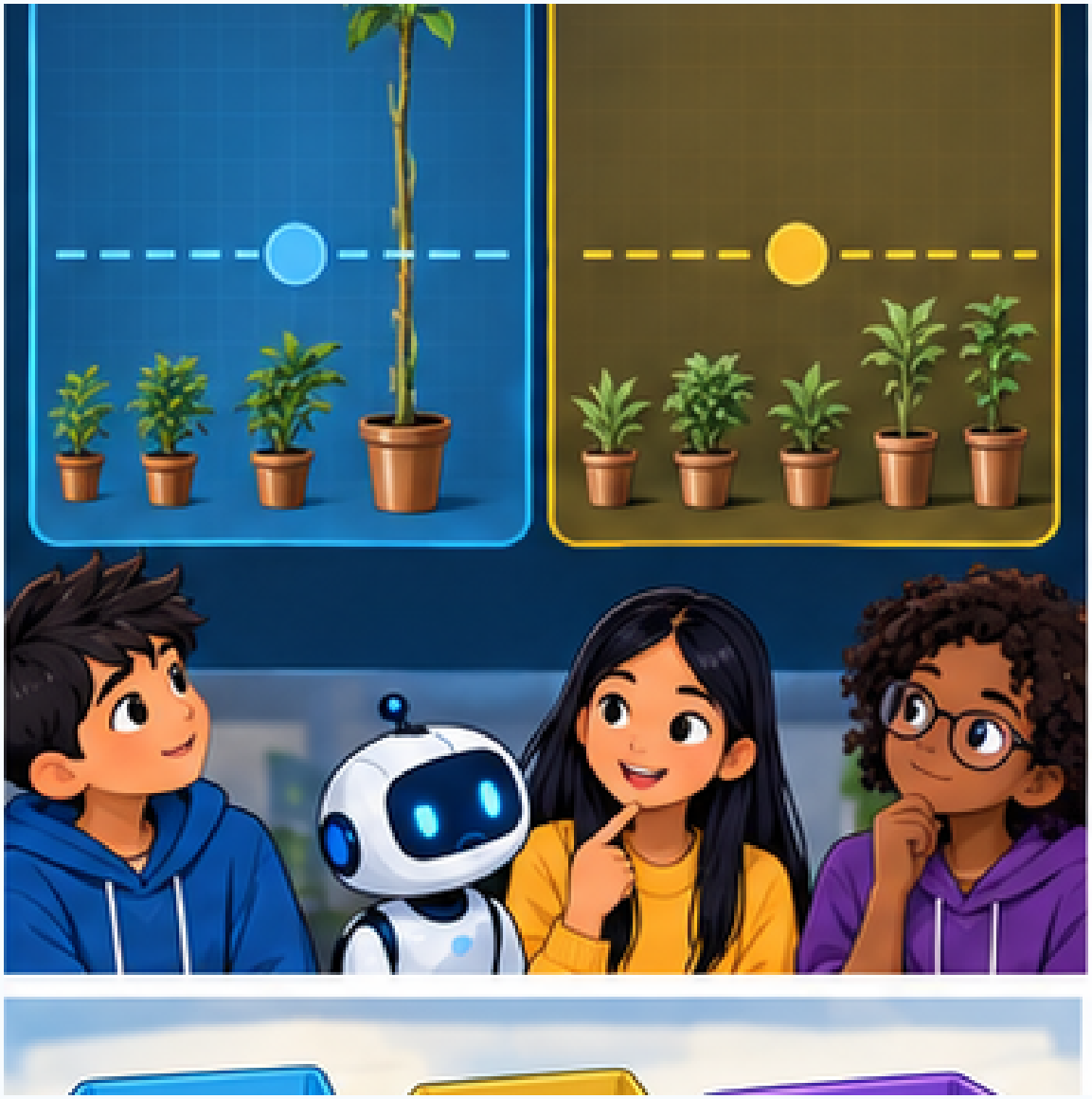
<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Mean and **median** are transparent regression baselines; **outlier**s affect them differently.
</details>



LEARNING TARGET For **training** targets [1,3,11], compute **mean** and **median constant baseline**s. Compare total absolute **training** error for each. Where must final performance be measured?

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Define the mean and the sorted middle before scoring

> A median for an even number of values needs two middle observations.

Complete constant-fitting contract

Before a complex regression model, fit a simple constant predictor. A **mean** baseline uses the **training** average. Sort the **training** targets for a **median**: use the middle value for an odd count, or the average of the two middle values for an even count. The methods ignore query features but provide a reproducible reference. Evaluation targets must not determine the constants.

Fictional numeric-target scope

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

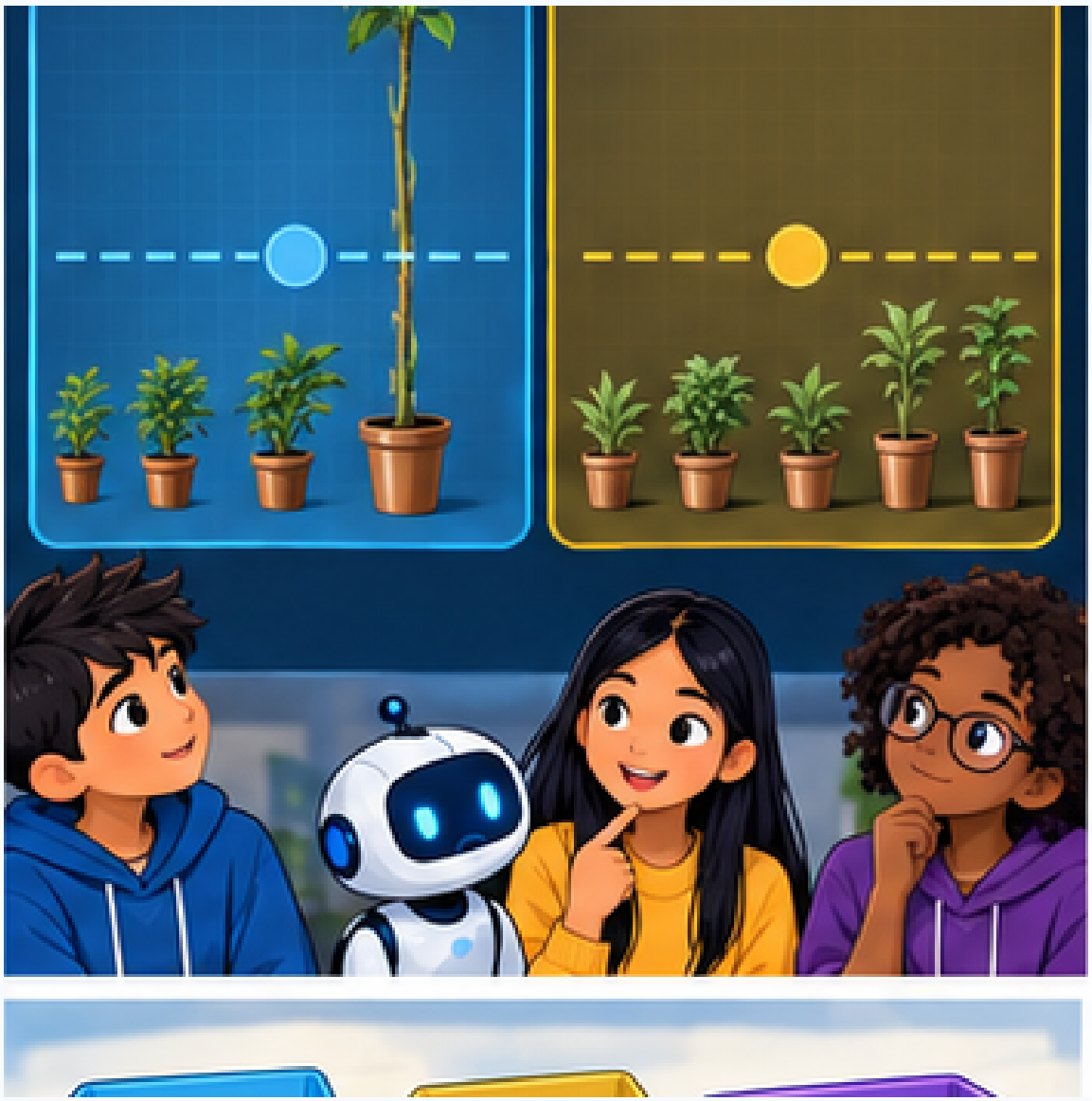
Earlier weeks supplied the model-building **process**; this week examines constant **prediction** baselines.

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<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.
</details>



MYSTERY The team must use **mean**, **median**, and baselines to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Fit two constants and score both on the same targets

> Equal validation MAE does not require equal predictions or errors.

Complete training and validation example

Training targets [2, 4, 6, 12] have **mean** 24/4=6 and **median** (4+6)/2=5. On separate targets [4, 8], the **mean** baseline predicts [6, 6] with **absolute error**s [2, 2], averaging 2.

The **median** baseline predicts [5, 5] with errors [1, 3], also averaging 2.

Optional original calculation program

```
from statistics import mean, median
training = [2, 4, 6, 12]
validation = [4, 8]
for name, value in [("mean", mean(training)),
                    ("median", median(training))]:
    mae = mean(abs(y - value) for y in validation)
    print(name, value, mae)
# Both validation MAEs are 2; constants differ.
```

Predict before running

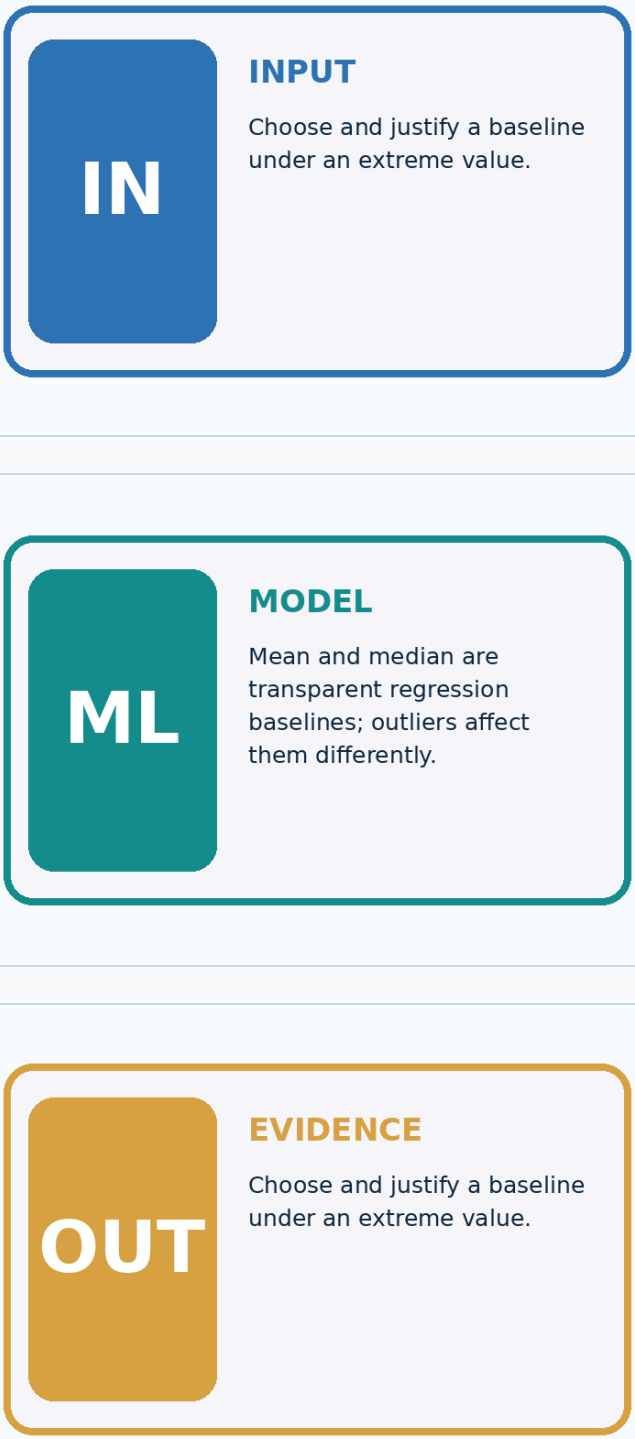
Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

▼ Earlier wording, alternatives and picture transcripts

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.



QUESTION Choose and justify a baseline under an extreme value.

MODEL IDEA **Mean** and **median** are transparent regression baselines; **outlier**s affect them differently.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Heights are 10, 11, 12, 13, and 54 cm. **Mean** = 100 ÷ 5 = 20 cm.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

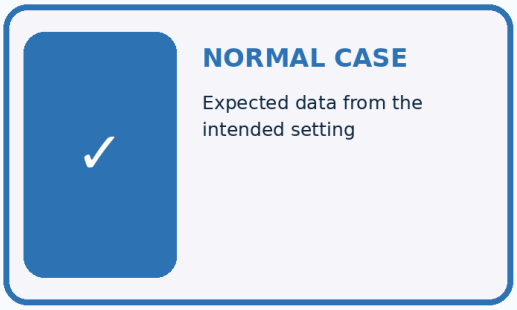
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

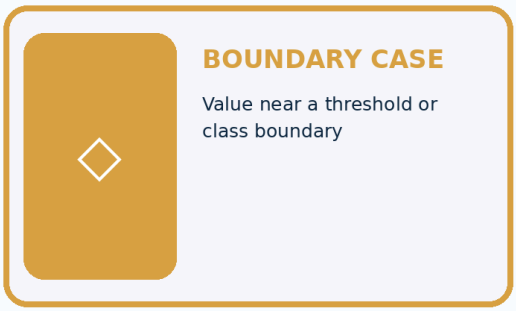
Compare training summaries with evidence about new cases

> A more stable training statistic is not automatically the best held-out predictor.

Four evaluation conditions



NORMAL Expected data and expected behavior.

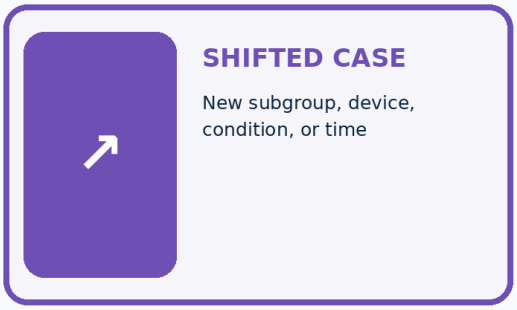


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Replace **training** 20 with 8 while keeping evaluation unchanged.

▼ **Check after predicting**

Prediction: Both fitted constants become5 cm and both evaluation MAEs remain3 cm.

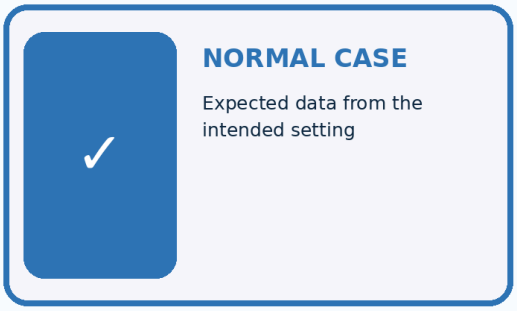
Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

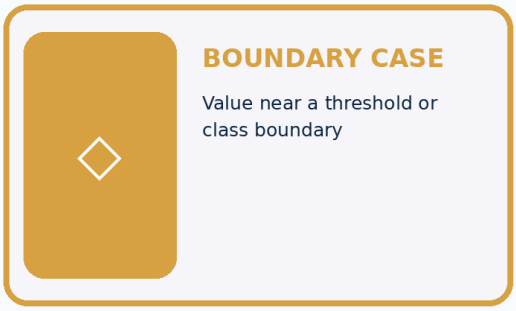
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

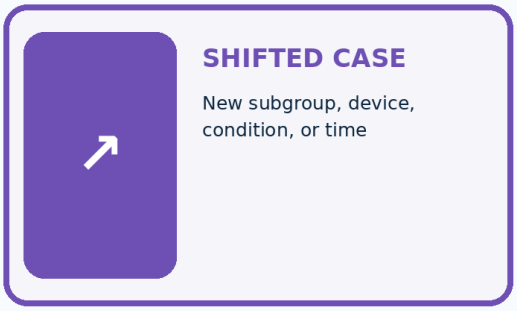
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the training statistic before the evaluation errors

> Keep the main and pictured training versions separate.

Complete pictured values and separate-fixture scope

Fit two constant height baselines and compare them on held-out examples.

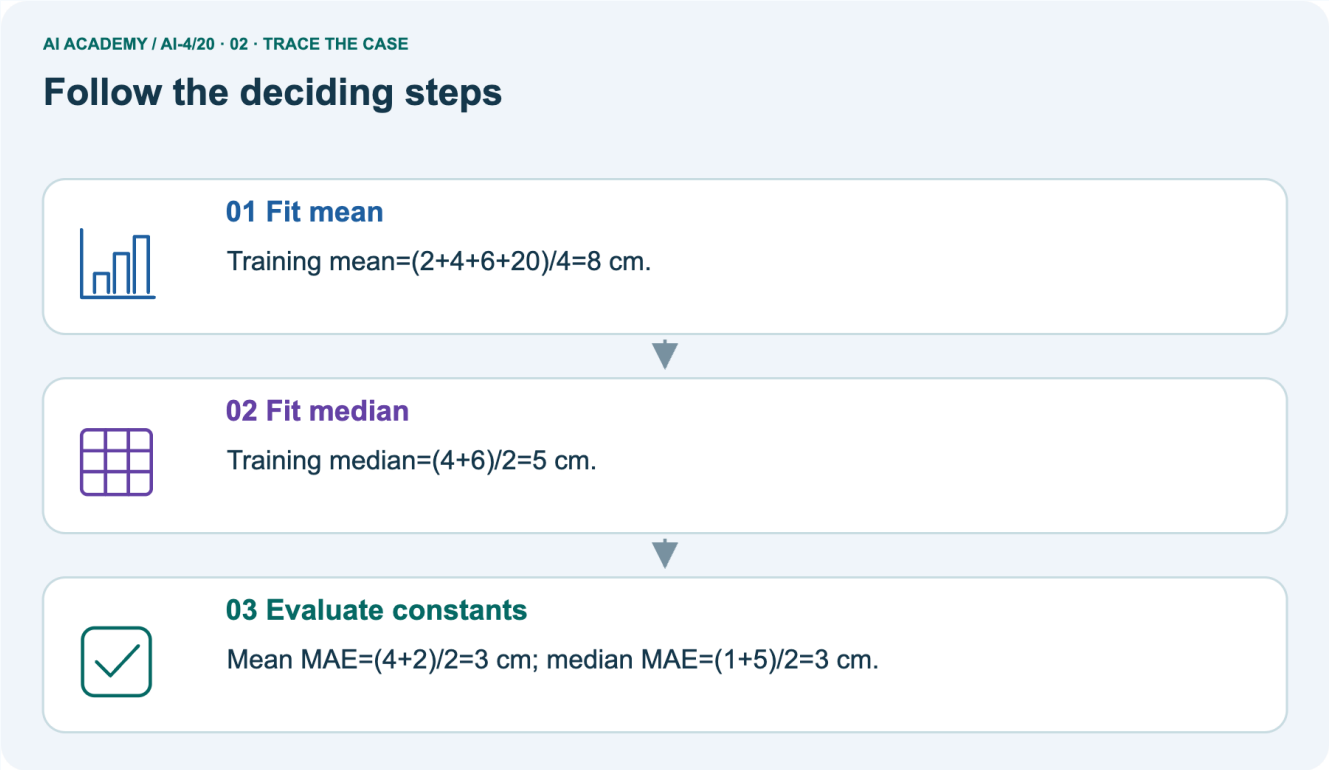
Training heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main training2,4,6,12gives mean6and median5;validation4,8gives errors[2,2]and[1,3],both MAE2cm. Picture training2,4,6,20gives mean8and median5;evaluation4,10gives errors[4,2]and[1,5],both MAE3cm. Replacing only pictured training20with8gives mean5and median5,both still evaluation MAE3cm. Keep the evaluation targets fixed.

This supplied change demonstrates sensitivity,not evidence that the extreme value was incorrect or permission to choose training edits from evaluation results.

Trace fitted constants and evaluation MAE



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit mean	Training mean=(2+4+6+20)/4=8 cm.
2. Fit median	Training median=(4+6)/2=5 cm.
3. Evaluate constants	Mean MAE=(4+2)/2=3 cm; median MAE=(1+5)/2=3 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

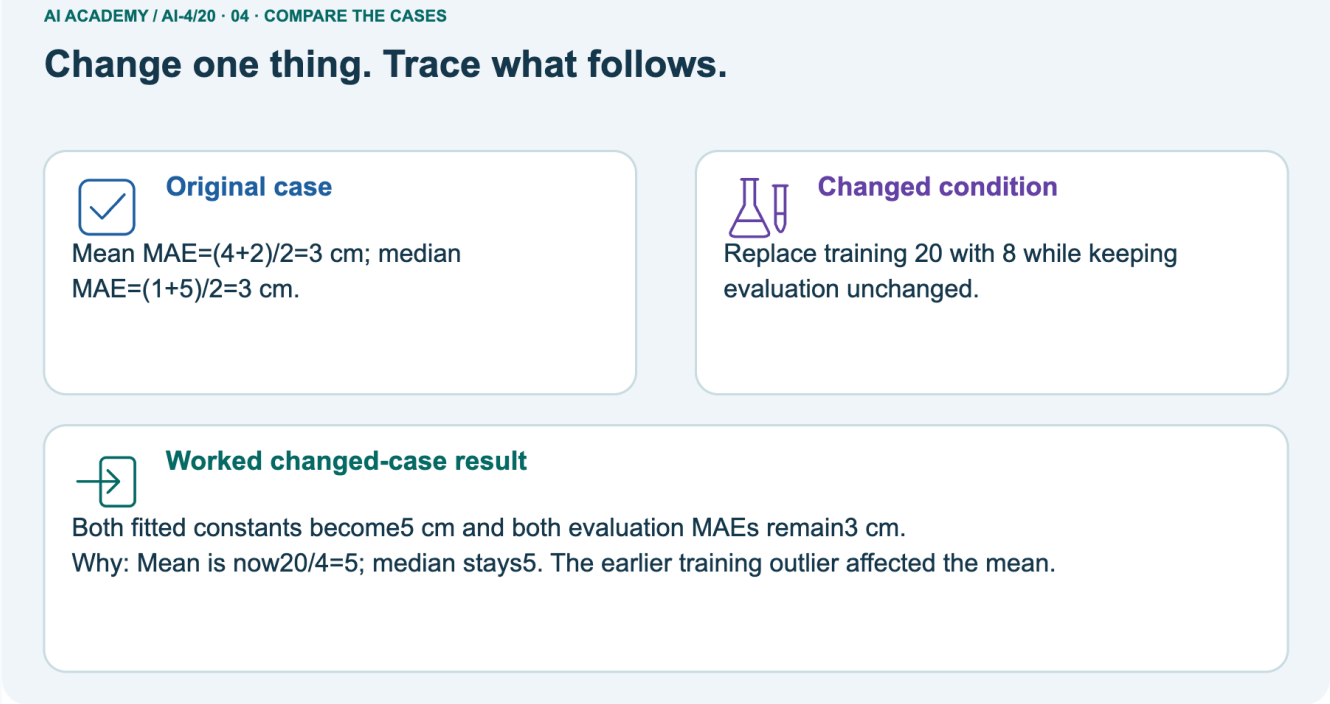
Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the pictured training-value replacement

- Change one thing: Replace **training** 20 with 8 while keeping evaluation unchanged. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain held-out and outlier limits

- **Mean** and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

- Read the labelled picture: Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main two-baseline score

Constant prediction baselines at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Training targets	Fit mean 6	Freeze constant
Training targets	Fit median 5	Freeze constant
Evaluation targets	Compare fixed predictions	Both MAE 2 here

Mean and median answer different training objectives. A mean minimizes total squared error among constants; a median minimizes total absolute error. This does not guarantee either wins on a new evaluation set. With extreme values, the mean can move much more than the median.

Calculate MAE for both baselines on [4, 8]. Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 20 • WORKED EVIDENCE

1. Heights are 10, 11, 12, 13, and 54 cm.
2. Mean = 100 ÷ 5 = 20 cm.
3. Median is 12 cm, the middle ordered value.
4. For typical-height absolute error, median is more robust; still report why 54 is present.

STEP 1 Heights are 10, 11, 12, 13, and 54 cm.

STEP 2 **Mean** = 100 ÷ 5 = 20 cm.

STEP 3 **Median** is 12 cm, the middle ordered value.

STEP 4 For typical-height **absolute error**, **median** is more robust; still report why 54 is present.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

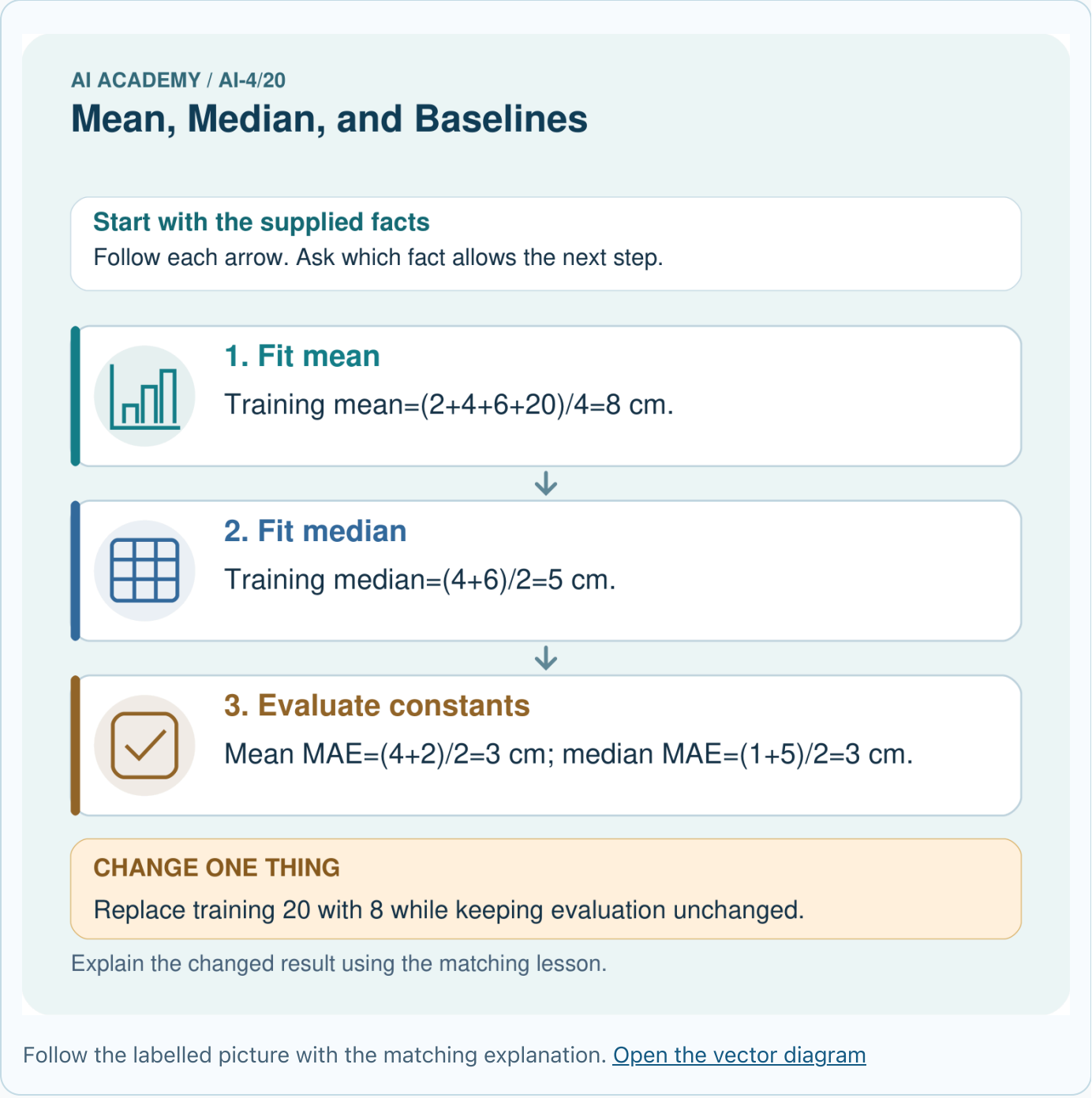
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Mean, Median, and Baselines

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a training summary mistaken for final performance

> Keep extreme values visible until their source has been checked.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE The **mean** is always the fairest and most scientific summary.

State interpretation and next-check limits

- For an even number of values, this course averages the two middle values to report one **median**. Keep a frozen baseline across comparisons and show its evaluation score.

  A model that fails to beat it may still reveal a data or metric issue worth investigating.

> **Why not predict the evaluation mean as the baseline?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The mean is always the fairest and most scientific summary. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise separate fitting and scoring with less help

> An inconvenient value needs source review rather than an automatic deletion.

Try supported W2

Use workbook W2 for the connected example “Constant **prediction** baselines”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Inspect training12becoming40

Replacing **training** 12 by 40 gives **mean** 13 and **median** 5. That change is a data change to record and investigate, not an instruction to delete an inconvenient **observation**.

Decide whether the extreme value is valid using the measurement policy and provenance.

Explain the evidence behind the final number

- Use the worked case for Constant **prediction** baselines. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured MAE tie

- Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the new independent training/evaluation set

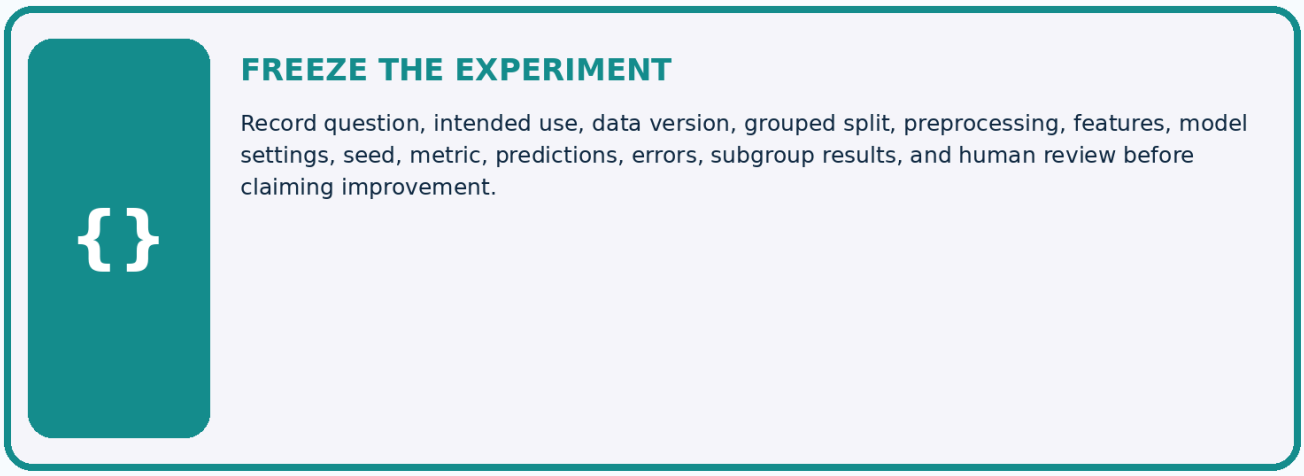
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Score separately

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete training-only Baseline Tournament

> The change record names which data changed and which evaluation remained fixed.

Complete original/changed fitting and scoring table

Baseline Tournament fixture	Training-only fitted constant	Same validation4,8cm:errors / MAE
Original training2,4,6,12cm;mean	6cm	[2,2] /2cm
Original training2,4,6,12cm;median	5cm	[1,3] /2cm
Changed training2,4,6,40cm;mean	13cm	[9,5] /7cm
Changed training2,4,6,40cm;median	5cm	[1,3] /2cm

Normal,single-value,missing and extreme probes

Complete baseline probe	Expected bounded check
Normal four-target training fixture	Fit mean6,median5before scoring separate4,8targets.

Complete baseline probe	Expected bounded check
Boundary one training value5cm	Mean and median both5; this is one observed training case,not a population guarantee.
Missing empty training or evaluation targets	No fitted statistic or no observed MAE; request data/report unavailable rather than invent0.
Extreme training12becomes40	Mean shifts6→13,median stays5;record source/version and investigate validity,do not infer corruption from size alone.

Fit,freeze,score and explain sensitivity

1. Fit both original constants without using validation targets, list every fixed prediction and compute its per-case absolute error. Compare the equal MAEs,then replay the supplied training-value change while leaving validation unchanged. Explain which statistic moved and why; mark the extreme value’s truth as unverified rather than silently removing it.

   Use the complete optional program for the same arithmetic,not a real dataset-quality decision. No complex regression model or final test is supplied. Keep the selected1,3,5,15training set for an uncoached first attempt.

Optional complete guarded tournament program

```
import math
from statistics import mean, median


def check_heights(values):
    if not values:
        raise ValueError('Need nonempty observed heights')
    if any(type(v) not in (int, float) or not math.isfinite(v) or v < 0 for v in values):
        raise ValueError('Need finite nonnegative heights in the declared unit')


def fit_constants(training):
    check_heights(training)
    return {'mean': mean(training), 'median': median(training)}
```

```
def evaluate(constants, targets):
    check_heights(targets)
    results = {}
    for name, value in constants.items():
        errors = [abs(y - value) for y in targets]
        results[name] = {'constant': value, 'absolute_errors': errors, 'mae':
mean(errors)}
    return results


training = [2, 4, 6, 12]
validation = [4, 8]
constants = fit_constants(training)
print(constants)
print(evaluate(constants, validation))
# Investigate a supplied training-value change; validation stays the same.
changed_constants = fit_constants([2, 4, 6, 40])
print(changed_constants)
print(evaluate(changed_constants, validation))
```

Keep evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

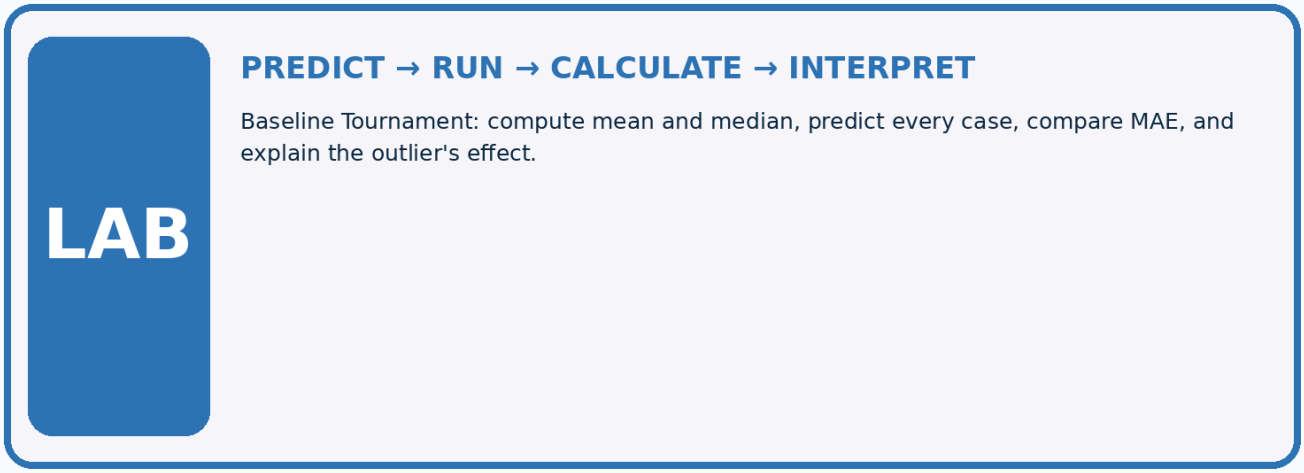
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Baseline Tournament: compute **mean** and **median**, predict every case, compare MAE, and explain the **outlier**'s effect.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: fit and score two new constant predictors

> Show training statistics and evaluation errors as separate stages.

Complete independent values and scope

Training [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

Use the selected training values1,3,5,15to fit both constants,then keep them fixed while scoring the supplied3,9targets. All values use the same numeric target/unit for this exercise. Neither the pictured training20nor the main training12belongs in this task. No final-test population or reliability claim is supplied.

First attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Apply independently

Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

For **training** heights 2, 3 and 10 cm, calculate **mean** and **median**. Which constant is less influenced by the high value?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result
</details>

Write or draw your reasoning and evidence.

Connect to your project

Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and **outlier** decision.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

For **training** targets [1,3,11], compute **mean** and **median constant baseline**s. Compare total absolute **training** error for each. Where must final performance be measured?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Keep the numeric reference method beside the project experiment

> The record must say what was fitted and where performance was measured.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and **outlier** decision.

For the numeric side experiment,save target definition/time/unit,training/evaluation source versions,fitted constant and method,and each held-out prediction/error. Keep outlier source questions and any supported correction with the raw value. Do not mix numeric MAE with classification accuracy or refit a constant on evaluation labels. A baseline comparison is conditional on these particular evaluation cases.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

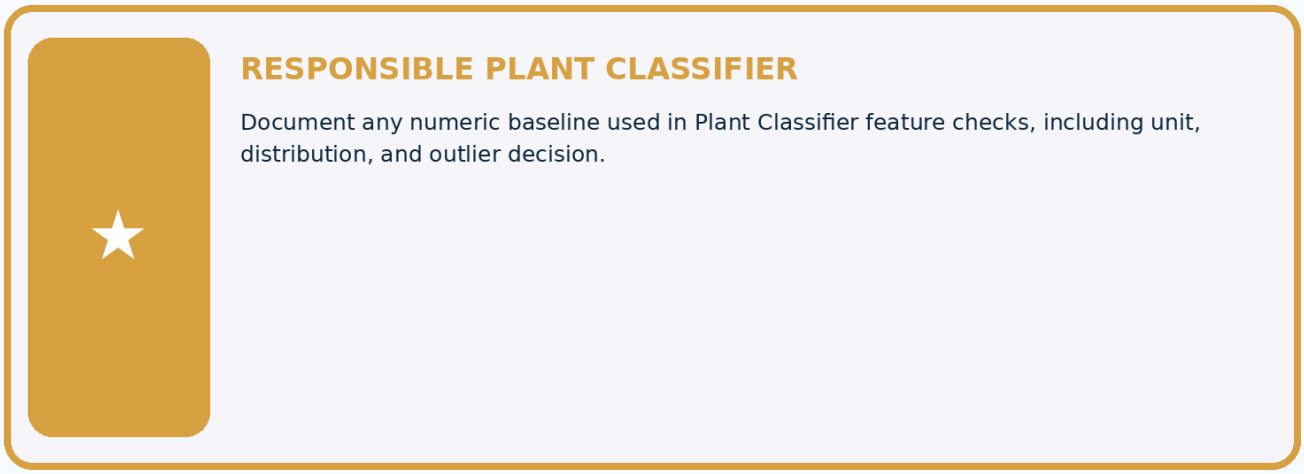
My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish by naming the training statistic and evaluation evidence

> Explain why one MAE tie does not make the two methods identical.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Fit **mean** and **median constant baseline**s on **training** targets

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
mean	sum divided by count	
median	middle ordered value	
outlier	value far from most observations	
location baseline	single typical value predicted for all cases	

Closer-explanation terms

Word	Meaning
Constant baseline	One fixed prediction for every query
Training mean	Sum of training targets divided by their count
Training median	Middle value of sorted training targets

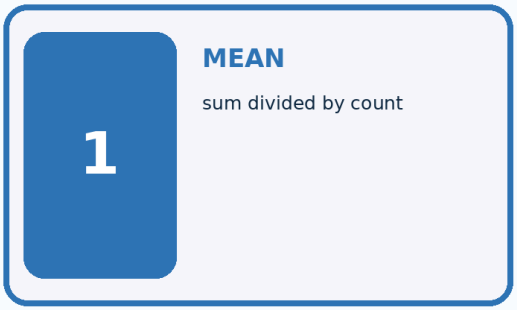
Word	Meaning
Absolute error	Magnitude of observed minus predicted

▼ Earlier wording, alternatives and picture transcripts

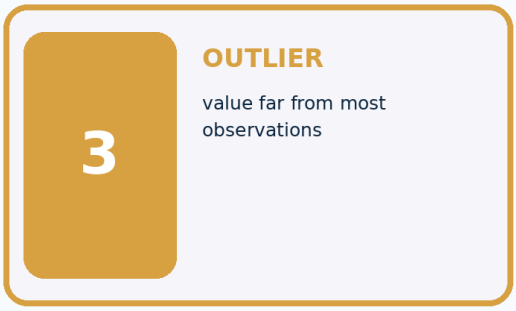
VOCABULARY

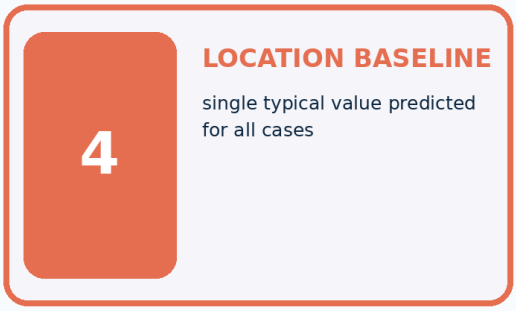
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR The **mean** is always the fairest and most scientific summary.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **mean**, **median**, **outlier**, **location baseline**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE The **mean** is always the fairest and most scientific summary.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

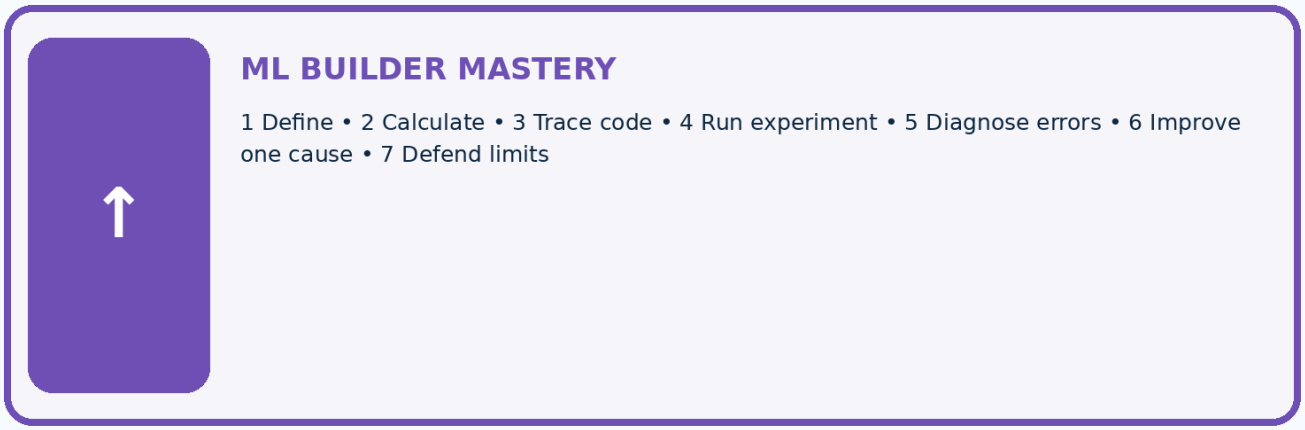
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh training1,3,5,15and evaluation3,9task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

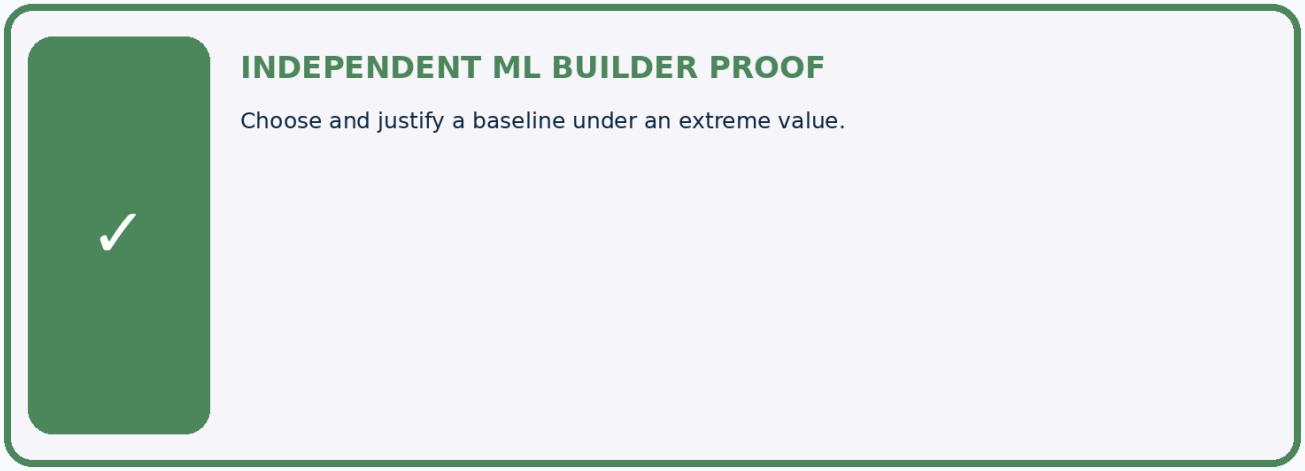
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Choose and justify a baseline under an extreme value.

NEXT BRIDGE Carry the experiment **evidence** into: **Absolute Error** and MAE.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Training heights are 2,4,12 cm. Compare constant **mean** and **median** baselines.

Compute both and **training absolute error**s.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Mean=6, **median**=4. **Mean** errors 4,2,6 total12; **median** errors 2,0,8 total10. **Median** minimises **absolute error** here.

Evidence or stage	Value or result
Observation A	2
Observation B	4
Observation C	12
Mean baseline	6
Median baseline	4

▼ Earlier wording, alternatives and picture transcripts

Week 20 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Mean and median baselines use the same observations

Training heights are 2,4,12 cm. Compare constant **mean** and **median** baselines.

Compute both and **training absolute error**s.

Mean=6, **median**=4. **Mean** errors 4,2,6 total12; **median** errors 2,0,8 total10. **Median** minimises **absolute error** here.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Fit mean and median constant baselines on training targets	Explain the deciding step, show a trace or test, and state one limit.
Compare constant predictions on separate records	Explain the deciding step, show a trace or test, and state one limit.
Keep fitted constants independent of evaluation answers	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Fit two constant height baselines and compare them on held-out examples.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Fit two constant height baselines and compare them on held-out examples.

- Which labelled step matches this detail? **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.
- 1. Fit **mean**
- 2. Fit **median**
- 3. Evaluate constants
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Evaluate constants **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Replace **training** 20 with 8 while keeping evaluation unchanged. Which conclusion is supported by the supplied facts?
- 1. Both fitted constants become5 cm and both evaluation MAEs remain3 cm.

- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Repair and keep the record

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric **prediction** from classification. Short practice: Which statistic reveals the size of errors without cancellation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score **mean** every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
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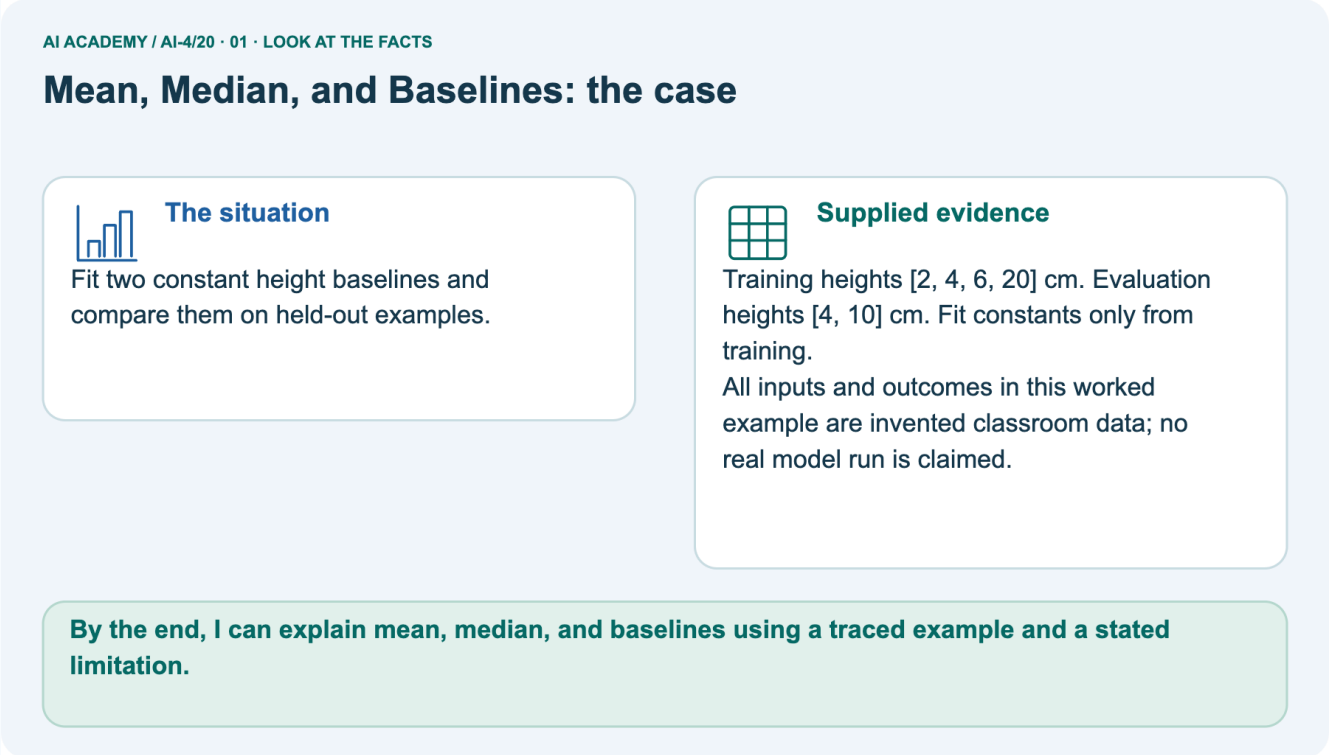
<details>
<summary>Reference · The six-picture journey</summary>
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REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Mean, Median, and Baselines: the case



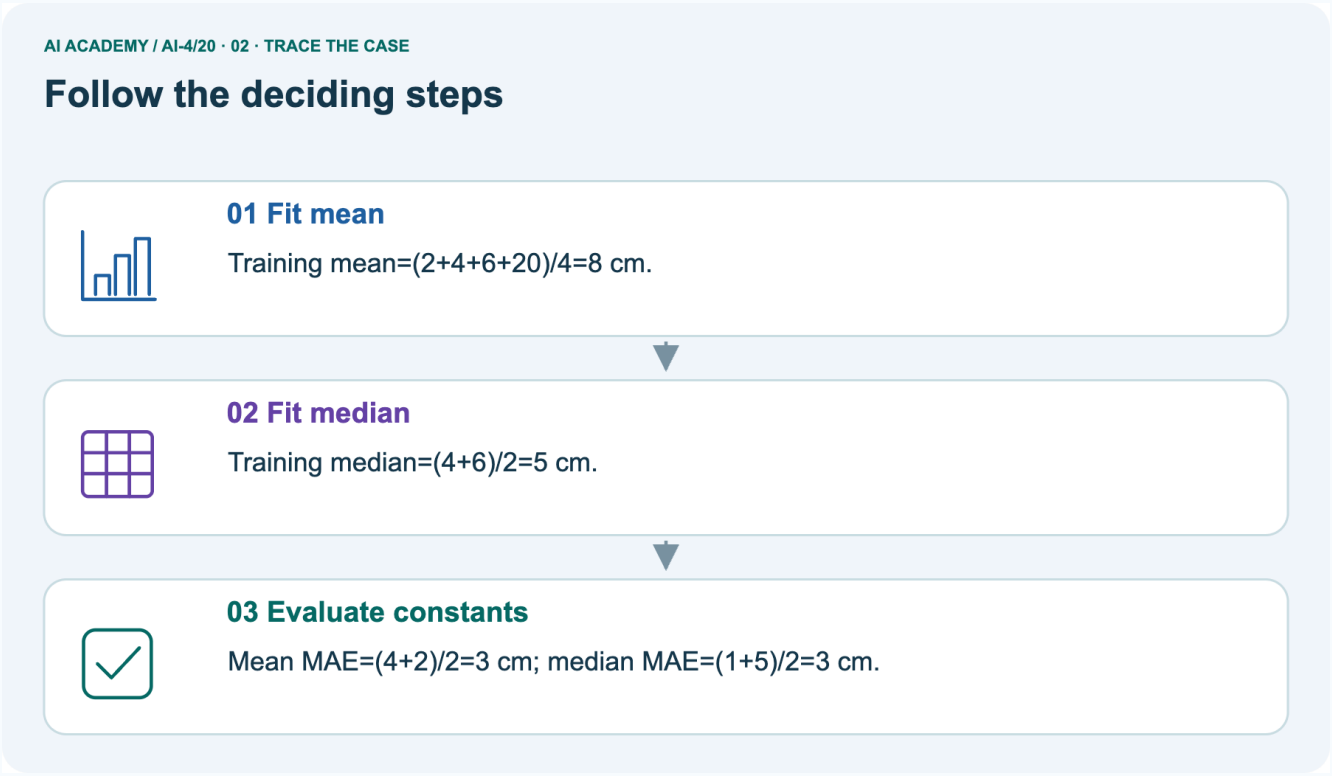
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit two constant height baselines and compare them on held-out examples. Supplied facts: **Training** heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

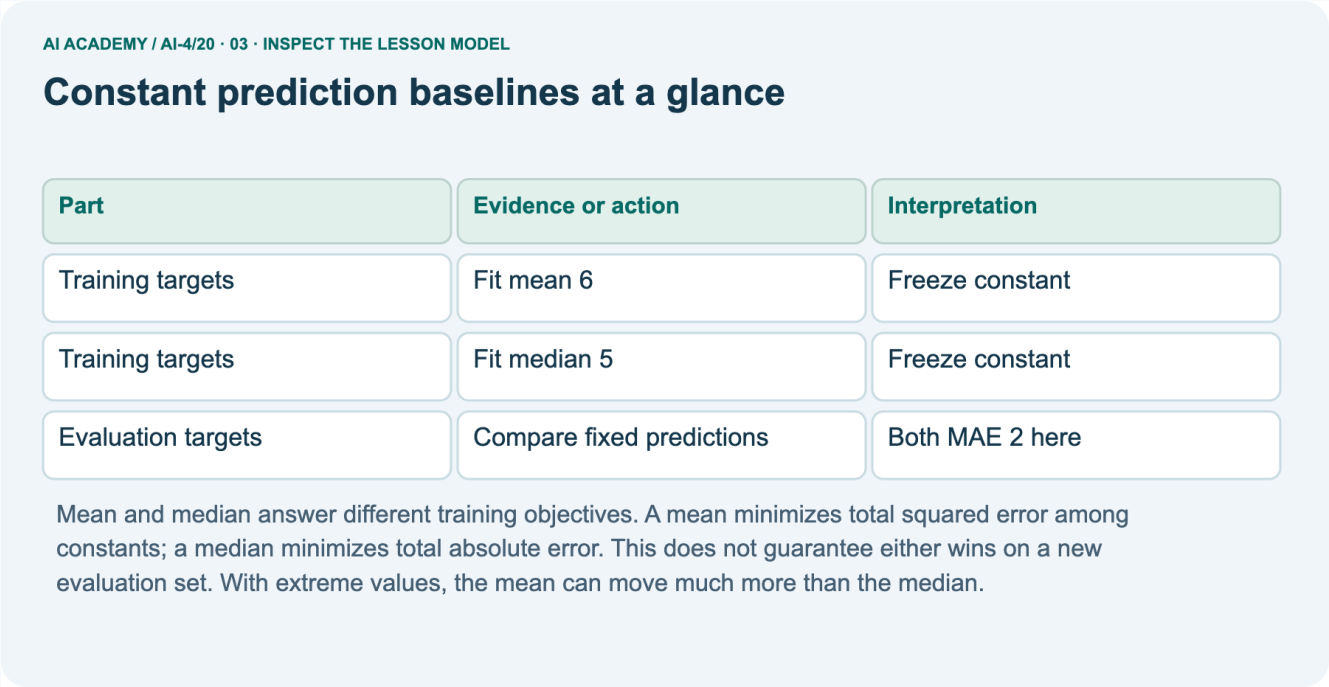


Follow the labelled picture with the matching explanation. Open the vector diagram

Fit **mean**: **Training mean**=(2+4+6+20)/4=8 cm. → Fit **median**: **Training median**=(4+6)/2=5 cm. → Evaluate constants: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

Point to each numbered stage and explain which recorded **input** it uses.

03. Constant prediction baselines at a glance

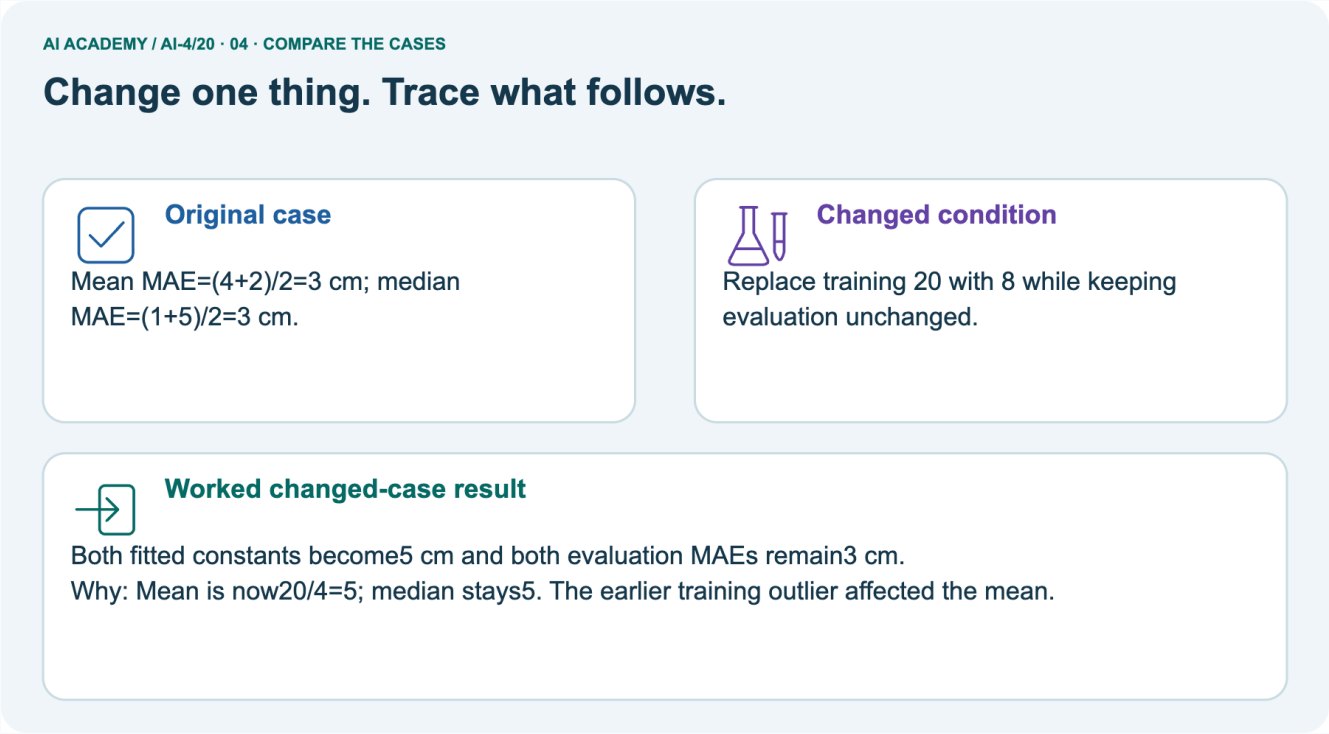


Follow the labelled picture with the matching explanation. Open the vector diagram

Constant **prediction** baselines at a glance. **Training** targets | Fit **mean** 6 | Freeze constant. **Training** targets | Fit **median** 5 | Freeze constant. Evaluation targets | Compare fixed **prediction**s | Both MAE 2 here

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



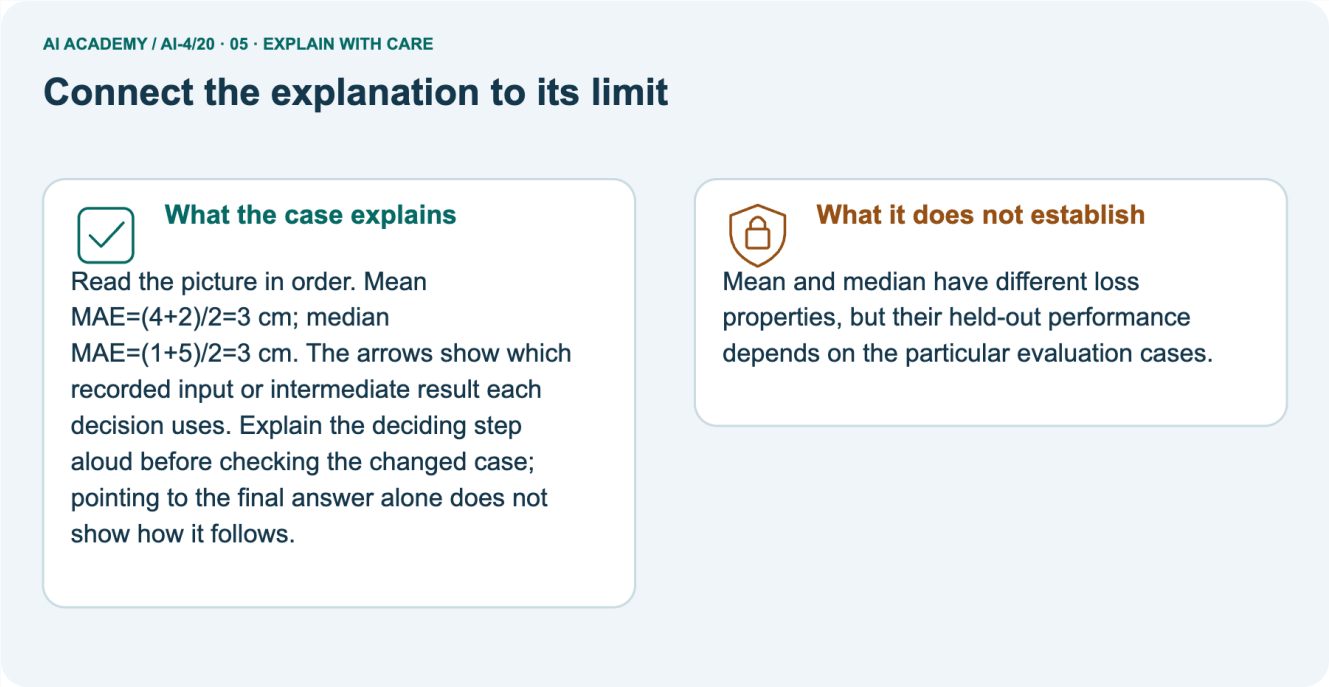
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. Changed condition: Replace **training** 20 with 8 while keeping evaluation unchanged.

Expected result: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

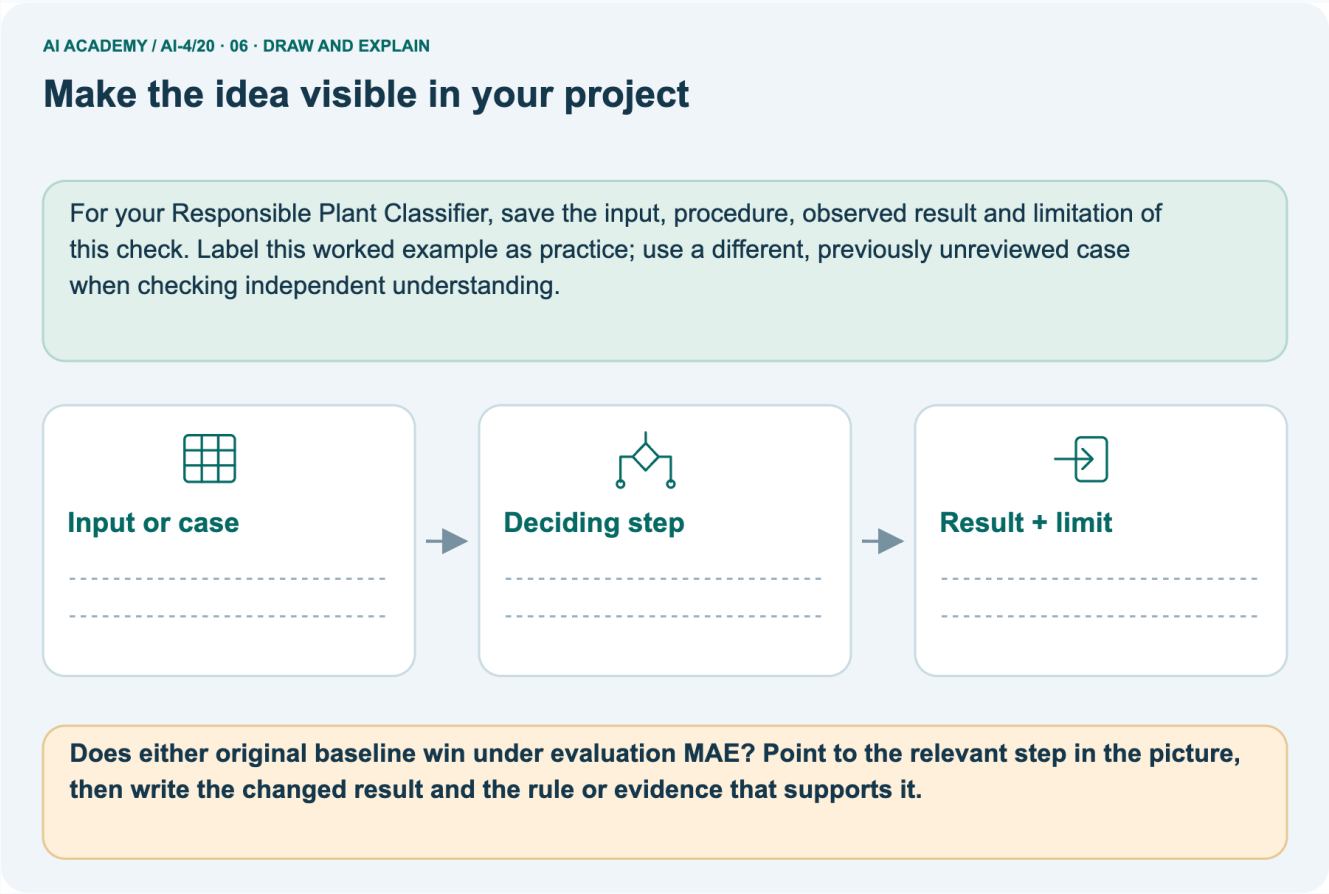
Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Mean** and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Mean, Median, and Baselines: the case

02. Follow the deciding steps

03. Constant prediction baselines at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.